COURSE OVERVIEW

MGMT 648 · Unit 1

Kerry Back

Thu Aug 27, 2026

Course topics

Two questions run through the term: whether a company should make a particular investment, and what a whole business is worth.

Capital budgeting

Statements to free cash flow, project cash flows, the decision rules, and simulation. Four units on September 12.

Company valuation

Discounted cash flow to a terminal value, then multiples of comparable firms as the cross-check. The two Thursdays in between.

Rates and portfolios

Mean-variance optimization, the cost of capital, bond pricing and duration, immunization. October 10.

Excel

Still the right tool for a tabular model that adds and subtracts, and the one your colleagues can open, change, and argue with.

Everything you present this term is a workbook someone else can drive.

AI

Everything else: fetching filings, estimating a beta, optimizing a portfolio, running ten thousand simulations.

It also builds the workbook — which is why half of tonight is about checking what it hands back.

Tools

Course materials

Everything is posted at mgmt648.kerryback.com.

Schedule

The home page. Each unit's slides and files sit beside its row, in HTML and PDF.

Syllabus

Grading, meeting times, and what to prepare. Nothing outside class is collected.

Free readings

Welch's *Corporate Finance*, Damodaran's spreadsheets, SEC EDGAR, and the learn-investments site. Nothing to buy.

EXCEL

What makes a workbook worth handing over

A table

Numbers computed somewhere else and pasted in. Change an assumption and it is quietly wrong — and nobody can challenge it, because there is nothing to change.

A model

Inputs in labeled cells, formulas that are real formulas. Change the rate and the sheet recalculates.

> Every graded number this term comes out of a model. A table cannot be defended, because a challenger has nothing to push on.

1. Every assumption gets its own labeled cell, in one block at the top
2. Formulas reference those cells, never a number typed inside the formula
3. The same formula runs across every period of a projection
4. Every hardcoded number carries its source beside it

> Write `=B5*(1+$B$6)`, not `=B5*1.05`. Source format: document, date, specific reference — “Company 10-K, FY2025, page 45, revenue note.”

Best practices — structure

Best practices — presentation

CONVENTION	RULE
Blue text	Hardcoded inputs — the cells a reader is meant to change
Black text	Every formula and calculation
Green text	Links to another sheet in the same workbook
Currency	`$#,##0`, with units in the header: Revenue ($mm)
Percentages	One decimal. Valuation multiples as `0.0x`
Negatives	Parentheses, (123), not −123

Zero formula errors on delivery: no `#REF!`, `#DIV/0!`, `#VALUE!`, or `#NAME?`.

Why the colors matter

The color convention is not decoration. It is how a reader knows, at a glance, what they are allowed to change.

For your reader

Blue says “this is an assumption, argue with it.” Black says “this follows from the assumptions, do not touch it.”

For you

A black cell that should have been blue is a number buried inside a formula, and that is the most common way a model goes quietly wrong.

AI

1. Go to lab648.kerryback.com
2. Username is your netid
3. Password is your student number, `S0...`

> A Linux container on a cloud server, billed to the school. Security on it is minimal — keep nothing personal there.

The lab

What is running there

The harness

Claude Code, connected to GLM 4.7 — a capable and relatively inexpensive model from Z.ai.

The software

Python and its scientific libraries, plus skills for financial data, spreadsheets, and slides.

The screen

File browser and terminal on the left; the prompt window at the foot of the right-hand panel.

Three things that will trip you up

Refresh

The file browser does not notice a new file on its own. Use the refresh button above it.

No Excel in the lab

Office is not installed. Double-clicking a workbook there does nothing.

Download to open

Workbooks come down to your laptop through the three-dots menu, and open in your own Excel.

AI AND CODING

The model only writes text

The model cannot open a file, run a calculation, or fetch a filing. It reads text and writes text. That is the entire capability.

What makes it useful is the harness around it. The model emits a request to use a tool; the harness runs the tool and reports the result back.

> The model never runs any of it.

1. The harness sends the prompt, the conversation, and the list of tools
2. The model answers: use tool X, with these arguments
3. The harness runs X and reports the result back
4. The model decides what comes next — another tool, or an answer for you

> On the lab, GLM writes a Python file, runs it at the terminal, reads the output, and revises. Four tool calls before you see anything.

The loop

What that produces

Excel workbooks

Written cell by cell with `openpyxl`: inputs as numbers, formulas as formulas.

Charts and figures

Any chart, any annotation, any label — specified in a sentence.

Word and PowerPoint

Real files with editable text, not pictures of them.

Data analysis

Filings from EDGAR, prices, the yield curve, factor returns — fetched, merged, analyzed.

Simulation and optimization

Ten thousand draws of an NPV, or a frontier over twenty assets.

Web apps

A page someone else can use, from a description of what it should do.

> Build an Excel workbook containing a mortgage amortization table. Put the loan amount, rate, and term in labeled input cells at the top. Include a chart of the amount going to interest and the amount going to principal each month.

> No mention of Python, and none needed. It writes the file cell by cell and hands you the workbook.

Ask for a workbook

THE WORKBOOK

Inputs in labeled cells, an outputs block below them, 360 rows of schedule.

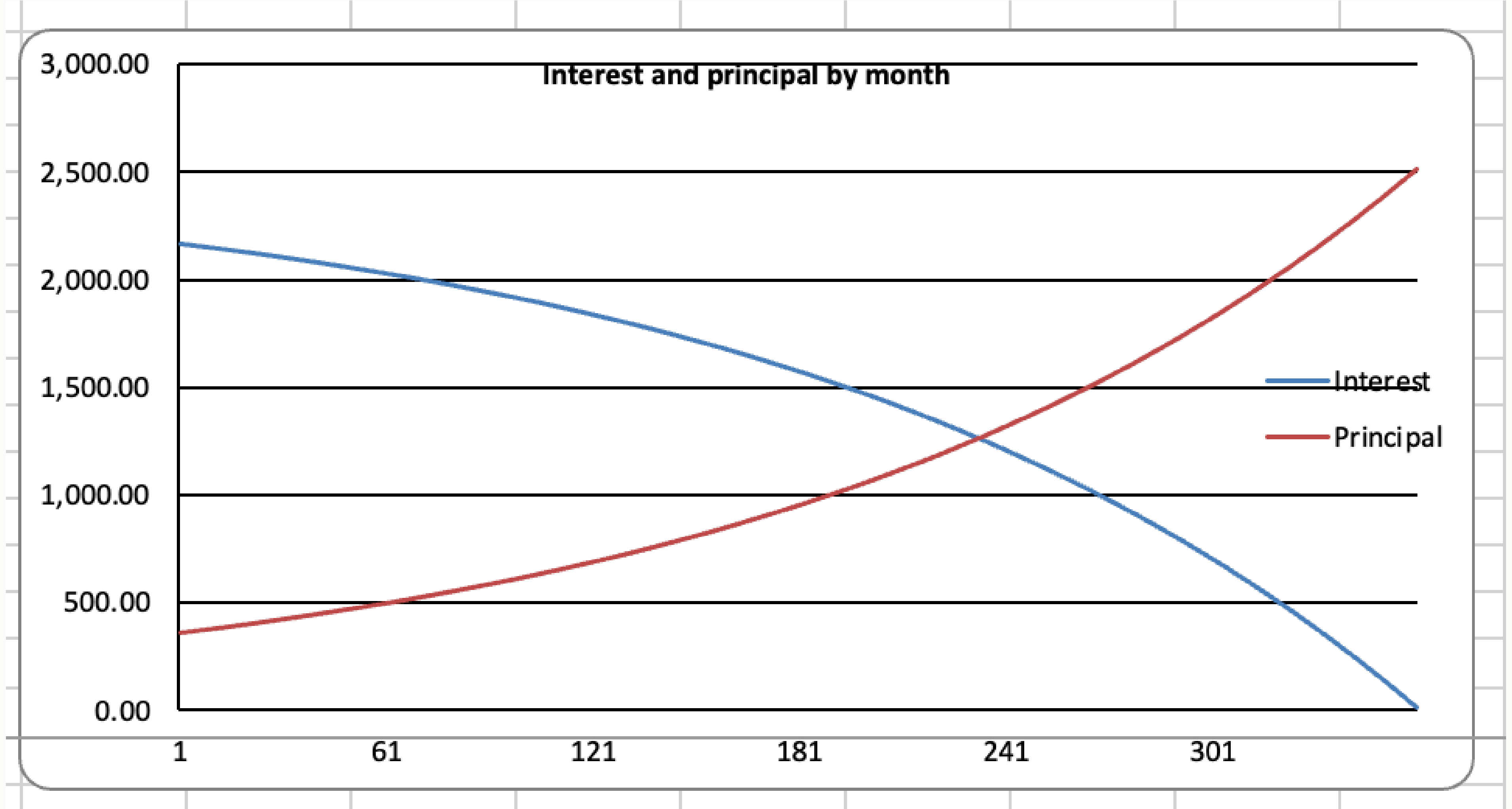
The selected payment cell holds `=$B$12`, not 2,528.27.

Change the rate and the whole sheet recalculates.

C18 | fx =$B$12

	A	B	C	D
1	**Mortgage Amortization**			
2				
3	**Inputs**			
4	Loan amount	**$400,000**		
5	Annual rate	**6.50%**		
6	Term (years)	**30**		
7	Payments per year	**12**		
8				
9	**Outputs**			
10	Periodic rate	0.5417%		
11	Total payments	360		
12	Monthly payment	**$2,528.27**		
13	Total interest	$510,178		
14				
15				
16	**Amortization schedule**			
17	**Month**	**Beginning balance**	**Payment**	**Interest**
18	1	400,000.00	2,528.27	2,166.
19	2	399,638.39	2,528.27	2,164.
20	3	399,274.83	2,528.27	2,162.
21	4	398,909.30	2,528.27	2,160.
22	5	398,541.78	2,528.27	2,158.
23	6	398,172.28	2,528.27	2,156.
24	7	397,800.77	2,528.27	2,154.
25	8	397,427.26	2,528.27	2,152.
26	9	397,051.71	2,528.27	2,150.
27	10	396,674.14	2,528.27	2,148.
28	11	396,294.52	2,528.27	2,146.

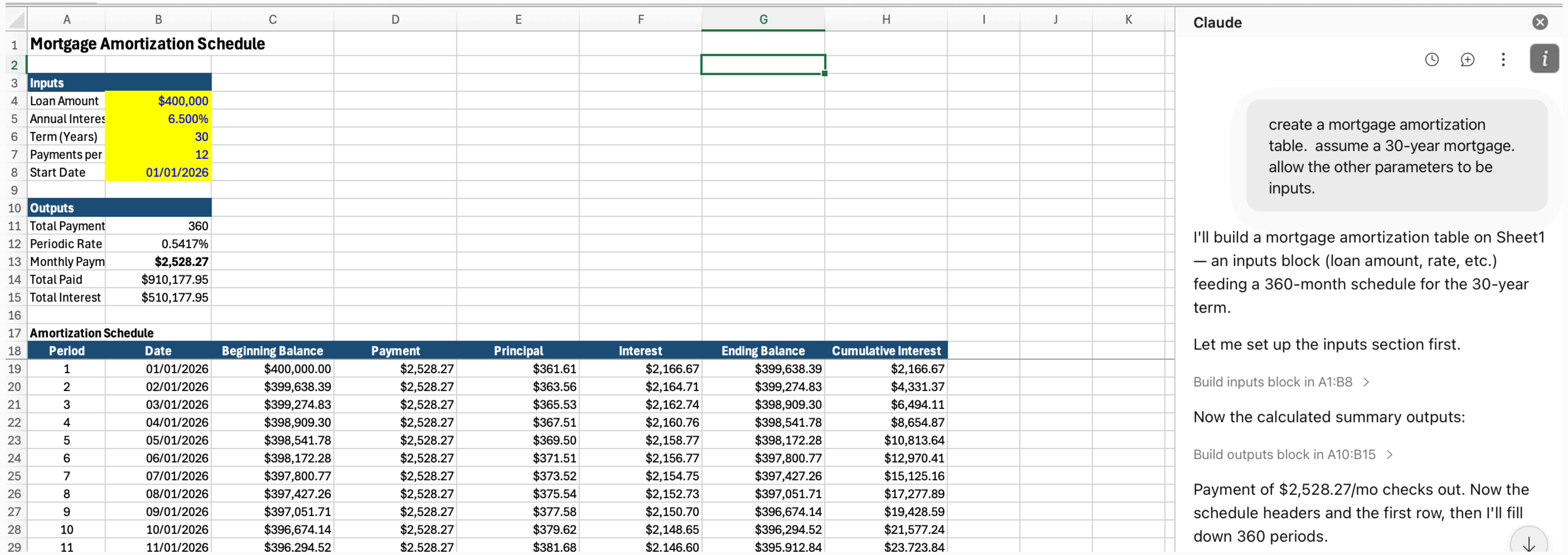
And the chart it asked for



> A chart object bound to the cells, not a picture pasted in. Change the rate and the crossover moves with it.

Claude inside Excel

There is also a chat panel in Excel itself, from `Home → Add-ins`. It sees the workbook already open, so you can ask about the sheet in front of you.



Mortgage Amortization Schedule

Inputs	
Loan Amount	$400,000
Annual Interes	6.500%
Term (Years)	30
Payments per	12
Start Date	01/01/2026
Outputs	
Total Payment	360
Periodic Rate	0.5417%
Monthly Paym	**$2,528.27**
Total Paid	$910,177.95
Total Interest	$510,177.95

Amortization Schedule

Period	Date	Beginning Balance	Payment	Principal	Interest	Ending Balance	Cumulative Interest
1	01/01/2026	$400,000.00	$2,528.27	$361.61	$2,166.67	$399,638.39	$2,166.67
2	02/01/2026	$399,638.39	$2,528.27	$363.56	$2,164.71	$399,274.83	$4,331.37
3	03/01/2026	$399,274.83	$2,528.27	$365.53	$2,162.74	$398,909.30	$6,494.11
4	04/01/2026	$398,909.30	$2,528.27	$367.51	$2,160.76	$398,541.78	$8,654.87
5	05/01/2026	$398,541.78	$2,528.27	$369.50	$2,158.77	$398,172.28	$10,813.64
6	06/01/2026	$398,172.28	$2,528.27	$371.51	$2,156.77	$397,800.77	$12,970.41
7	07/01/2026	$397,800.77	$2,528.27	$373.52	$2,154.75	$397,427.26	$15,125.16
8	08/01/2026	$397,427.26	$2,528.27	$375.54	$2,152.73	$397,051.71	$17,277.89
9	09/01/2026	$397,051.71	$2,528.27	$377.58	$2,150.70	$396,674.14	$19,428.59
10	10/01/2026	$396,674.14	$2,528.27	$379.62	$2,148.65	$396,294.52	$21,577.24
11	11/01/2026	$396,294.52	$2,528.27	$381.68	$2,146.60	$395,912.84	$23,723.84

> It needs a paid Anthropic account and your own copy of Excel, so it is not the route this course runs on. Worth knowing it exists.

BEST PRACTICES FOR USING AI

Plan before you act

Suppose it told you last month that method A beats method B. Today you ask it to do the task — and it reaches for B, because “A beats B” is not in *this* conversation and B appeared more often in training.

Planning pulls the knowledge in

Plan how you will do this before you start. The plan puts the relevant considerations into the conversation, and it answers in light of those rather than from base rates.

Planning breaks the work up

Steps you can check one at a time. Failures stay small and local instead of surfacing at the end, when everything downstream is already wrong.

The problem

Ambiguity does not produce a question. It produces a confident answer to a question you did not ask, and you will not notice until the work is done.

The fix

Tell me what you are about to do and what you are assuming.

Correct it in one sentence. Then let it go.

Confirm it understood

Numbers need code

To a next-token predictor, 47.3% is exactly as plausible as 52.1%.

Where it goes wrong

Mental arithmetic, rankings, comparisons, aggregations. The wrong number arrives with complete confidence.

The fix

Have it write Python and actually run it. Arithmetic stops being prediction and becomes computation, and you can read the code afterward.

The workbook says the payment on $400,000 at 6.5% for 30 years is $2,528.27, because `PMT` ran in a cell.

Ask the same question in prose and you get a number shaped like that one — close enough to look right, wrong enough to matter.

> Worth doing live: the same question twice, once in prose and once with "compute it with Python." Post both numbers.

Ask it without code

What not to do

The 2022 prompting folklore is now useless or actively harmful.

SKIP	WHY
“Think step by step”	Reasoning models already do, in hidden tokens
“Take a deep breath”	Measurably useless now
“CRITICAL! NEVER EVER...”	Overtriggers, and gets worse results
Stacked personas	One helps the tone; three does nothing

Say what you want, say what done looks like, say what to avoid.

CRITIQUE AND VALIDATE

Never ask whether your idea is good

Training on human feedback teaches that agreement earns high ratings. Ask “is this valuation sound?” and it will find reasons that it is.

> Invert it. *What is the strongest argument that this project should be rejected?* Better still, name the skeptic — a board member, a lender, the division head whose budget this comes out of.

Why a second instance

A model defends its own output. A fresh conversation, told to attack the work rather than summarize it, has nothing to defend.

What to tell it

Find every factual error, logical gap, and unsupported assumption in this model. Do not summarize it — find flaws.

Scale the checking to what the number is worth. A cost of capital you will defend for six minutes deserves a second opinion.

Set a critic on it

Three ways it goes wrong

Think about a human assistant, and ask where they could have gone wrong.

Understanding

They may not have understood what you wanted.

Decisions

They made choices you never discussed, and made them differently than you would have.

Bugs

Their spreadsheet had a subtle error, so it did not do what they thought it did.

Three questions, every time

1. *Explain to me what you did and how you did it.*
2. *Report every way my instructions were ambiguous and you made a decision we have not discussed.*
3. *Work one case through by hand so I can verify your number.*

> It is an infinitely patient colleague with no memory. You can probe far harder than you would with a person, and it will not be offended.

1. Does it foot? Do the parts sum to the total, and is the row count what you expected?
2. Is one number right? Trace a single cell back to the filing by hand.
3. Is it the right size? Say what you expect *before* you look.

> Afterwards, every number looks reasonable. That is why the prediction has to come first.

Then check the number itself

Facts are not code

Citations, filing dates, regulations, and tax rates are stated exactly the same way whether they are real or invented. Confidence is not correlated with accuracy.

The risk runs the wrong direction: the more obscure the item, the thinner the training data, and the likelier it is fabricated.

> Make it search before answering, and ask for the source. Then open the source.

Regenerating looks free

It is not. You get different choices each time — a different definition of the same variable — so two scripts that ought to agree quietly do not.

And bugs come back

A bug you found and fixed reappears in the new version, because nothing about the fix was ever written down.

Keep the code that worked

THE COURSE AHEAD

How a session works

Before

Watch the unit video, about twelve minutes, and compute one number for your company or the case. Forty-five minutes, none of it graded.

During

We collect the numbers, put the range on screen, and argue about the outliers. Then you build. The Saturdays are almost entirely keyboard time.

Graded

Participation, three live presentations, and serving as challenger. Nothing you do outside the room is collected.

1. Sign in to the lab tonight and build the amortization workbook
2. Watch the Unit 2 video, or read its transcript
3. Bring your company’s most recent EBIT and effective tax rate, from the filing

> September 12 runs 2:00 to 6:30, in person. Bring a laptop with Excel on it.

Before September 12

THE HABIT

If a number matters, it came from code that ran, and you checked it against something computed a different way.